

## Discussion 11: Particle physics reactions with 4-vector kinematics – Solutions

Names:

Section:

This week we will practice using 4-vectors, energy-momentum conservation, and the invariance of the relativistic dot product to solve problems in relativistic kinematics. Far from being artificial examples, these are the *most common* use of relativity in particle physics, and also have applications in nuclear physics as well. We will use natural units ( $c = 1$ ) throughout. To save some space, here is another convenient notation for the dot product of a 4-vector with itself:

$$p^2 \equiv p \cdot p \equiv p_\mu p^\mu \quad (1)$$

You've seen the latter two expressions before; I've avoided the first one in class because it can possibly be confused with the  $y$ -component of the 4-vector itself, but we will never be dealing with individual spatial components in this discussion. So you can always read  $p^2$  as “ $p$ -squared.”

### Part 1: Particle decays revisited

In lecture we studied the example of pion decay,  $\pi^+ \rightarrow \mu^+ + \nu$ . Let's generalize this example to the following situation: a particle of mass  $M$  decays to two particles with masses  $m_1$  and  $m_2$ .

- 1.1. Using energy-momentum conservation and the invariant dot product, find the energy of  $m_1$  in the rest frame of  $M$ . *Hint: this derivation proceeds exactly like the one in lecture, except one of the dot products will be  $m_2^2$  instead of 0.*

Let  $p$  denote a four-momentum vector and let  $\vec{p}$  denote a 3-momentum vector. Working in the rest frame of the  $M$  mass, the initial four-momentum of the mass  $M$ , and the final four-momenta of mass  $m_1$  and  $m_2$  respectively are:

$$p_M = \begin{pmatrix} M \\ \vec{0} \end{pmatrix}, \quad p_{m_1} = \begin{pmatrix} E_{m_1} \\ \vec{p}_{m_1} \end{pmatrix}, \quad \text{and} \quad p_{m_2} = \begin{pmatrix} E_{m_2} \\ \vec{p}_{m_2} \end{pmatrix}$$

By four-momentum conservation, we have that

$$p_i = p_f \implies p_M = p_{m_1} + p_{m_2}$$

We can rearrange this and use the invariance of the dot product (stated in Eq. 1) to write:

$$\begin{aligned} (p_{m_2})^2 &= (p_M - p_{m_1})^2 = p_M^2 - 2p_M \cdot p_{m_1} + p_{m_1}^2 \\ m_2^2 &= M^2 - 2ME_{m_1} + m_1^2 \end{aligned}$$

where we use  $p_M \cdot p_{m_1} = ME_{m_1}$ . We can now solve for  $E_{m_1}$  to conclude that

$$E_{m_1} = \frac{M^2 + m_1^2 - m_2^2}{2M}$$

- 1.2. Now do the same to find the energy of  $m_2$ . Verify that the sum of the final-state energies equals  $M$ ; this is a good check that you are applying energy conservation correctly!

We follow similar steps as before, and arrive at

$$(p_{m_1})^2 = (p_M - p_{m_2})^2 = p_M^2 - 2p_M \cdot p_{m_2} + p_{m_2}^2$$

$$m_1^2 = M^2 - 2ME_{m_2} + m_2^2$$

where we used  $p_M \cdot p_{m_2} = ME_{m_2}$ . We can now solve for  $E_{m_2}$  to conclude that

$$E_{m_2} = \frac{M^2 + m_2^2 - m_1^2}{2M}$$

We check the sum of the two energies to verify that the sum of the final-state energies is  $M$ :

$$E_{m_1} + E_{m_2} = \frac{M^2 + m_1^2 - m_2^2}{2M} + \frac{M^2 + m_2^2 - m_1^2}{2M} = \frac{2M^2}{2M} = M$$

- 1.3. Set  $m_2 = 0$  and verify that you recover the result for the energy of the massless neutrino we studied in lecture. Then set  $m_1 = m_2 = 0$  and re-derive your result for  $\pi^0$  decaying to massless photons from problem 3.4 of Discussion 8.

We set  $m_2 = 0$  in our equation for the energy of mass 2 (which is analogous to the neutrino):

$$E_{m_2} = \frac{M^2 + m_2^2 - m_1^2}{2M} = \frac{M^2 - m_1^2}{2M}$$

which is the result we found in lecture. Now, consider the process  $\pi^0 \rightarrow \gamma + \gamma$  that we studied in problem 3.4 in discussion 8. Our mass 2 is analogous to one of the photons. We set  $m_1 = m_2 = 0$  and we use our equation for the energy of particle 2, noting that  $M = m_\pi$ :

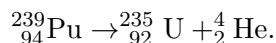
$$E_{m_2} = \frac{M^2 + m_2^2 - m_1^2}{2M} \rightarrow \frac{M^2}{2M} = \frac{m_\pi}{2}$$

which is the result we found previously.

- 1.4. Prove that  $M$  can only decay into  $m_1$  and  $m_2$  if  $M \geq m_1 + m_2$ . Decays into heavier final-state particles would violate conservation of energy!

Masses 1 and 2 have the minimum energy if they both come out of the process at rest. In this case, they'd each have an energy of  $\frac{M}{2}$  from conservation of energy and momentum. If they were any heavier, they would have to have negative energy, which is impossible. Hence, we conclude that  $M \geq m_1 + m_2$ .

- 1.5. An important application of these calculations is in nuclear decay. A plutonium nucleus can undergo alpha decay:



The mass of the  ${}^{239}_{94}\text{Pu}$  nucleus is  $M = 222.676$  GeV (where  $1 \text{ GeV} = 10^9 \text{ eV}$ ), and the masses of the daughter particles are  $m_1 = 218.942$  GeV for  ${}^{235}_{92}\text{U}$  and  $m_2 = 3.727$  GeV for  ${}^4_2\text{He}$ . (For historical reasons, the helium nucleus  ${}^4_2\text{He}$  is also known as an alpha particle.) Compute the

*kinetic* energy of each of the decay products,  $T = E - m$ . What fraction of the rest energy of the plutonium nucleus is converted to kinetic energy in this decay? This is  $E = mc^2$  in action!

We note that the final energy of our system comes from the rest mass of the particles plus their kinetic energy. The kinetic energy  $T$  can be calculated from subtracting the rest masses of the final products from the rest mass of the initial nucleus:

$$T = m_{\text{Pu}} - m_{\text{U}} - m_{\text{He}} = (222.676 - 218.942 - 3.727) \text{ GeV} = 7 \text{ MeV}.$$

This kinetic energy is about 0.003% of the rest energy of the Pu nucleus. Alternatively, we can use the energies computed in problems 1.1 and 1.2 and just subtract the masses to find the individual kinetic energies:

$$T_1 = E_{m_1} - m_1 = \frac{M^2 + m_1^2 - m_2^2 - 2Mm_1}{2M} = \frac{(M - m_1)^2 - m_2^2}{2M}$$

$$T_2 = E_{m_2} - m_2 = \frac{M^2 + m_2^2 - m_1^2 - 2Mm_2}{2M} = \frac{(M - m_2)^2 - m_1^2}{2M}$$

Plugging in numbers,  $T_1 = 0.11 \text{ MeV}$  and  $T_2 = 6.88 \text{ MeV}$ , which add to  $T = T_1 + T_2 = 6.99 \text{ MeV}$ , which is the same as we found above up to rounding error. (The fact that the heavier nucleus has less kinetic energy makes sense by conservation of momentum, which requires the heavier particle to move with slower velocity.)

## Part 2: Invariant mass

Most elementary particles (besides the familiar proton, neutron, and electron) are extremely short-lived. Therefore, it is often not possible to observe them directly, and instead we look for their decay products and try to reconstruct the presence of the original particle. A very common technique to do so is by considering the *invariant mass* of a set of decay products. In the case where the parent particle decays to two daughter particles (as in Part 1 above), with 4-momenta  $k_1$  and  $k_2$ , the invariant mass is defined as

$$m_{12}^2 = (k_1 + k_2)^2. \quad (2)$$

- 2.1. If  $k_1$  and  $k_2$  came from the decay of a particle of mass  $M$ , explain why  $m_{12}^2 = M^2$ , and justify the use of the term “invariant.”

In the rest frame of the initial particle, the 4-momentum can be written

$$p = (M, 0, 0, 0).$$

So by 4-momentum conservation

$$p^2 = (k_1 + k_2)^2 \implies M^2 = m_{12}^2.$$

The use of the term “invariant mass” is fitting because we defined it as  $m_{12}^2 = (k_1 + k_2)^2$ ; the relativistic dot product is invariant between frames, and its value is the (squared) mass of the parent particle.

- 2.2. Since the invariant mass is invariant, we can compute its value in any frame. Suppose that  $k_1$  and  $k_2$  have the same mass  $m$ , and in the laboratory frame,  $k_1$  and  $k_2$  have energies  $E_1$  and  $E_2$ , respectively. Compute  $m_{12}^2$  as a function of  $E_1$ ,  $E_2$ ,  $m$ , and  $\cos \theta_{12}$ , the angle between the 3-momenta of  $k_1$  and  $k_2$ .

Define:

$$k_1 = (E_1, \vec{p}_1), \quad k_2 = (E_2, \vec{p}_2)$$

where

$$|\vec{p}_{1,2}| = \sqrt{E_{1,2}^2 - m^2}.$$

Then we can compute the invariant mass

$$\begin{aligned} m_{12}^2 &= (k_1 + k_2)^2 \\ &= (E_1 + E_2, \vec{p}_1 + \vec{p}_2)^2 \\ &= (E_1 + E_2)^2 - (\vec{p}_1 + \vec{p}_2) \cdot (\vec{p}_1 + \vec{p}_2) \\ &= E_1^2 + E_2^2 + 2E_1E_2 - |\vec{p}_1|^2 - |\vec{p}_2|^2 - 2|\vec{p}_1||\vec{p}_2|\cos \theta_{12} \\ &= 2(E_1E_2 + m^2 - \sqrt{E_1^2 - m^2}\sqrt{E_2^2 - m^2}\cos \theta_{12}) \end{aligned}$$

Alternatively, we can compute the invariant mass as

$$\begin{aligned} m_{12}^2 &= (k_1 + k_2)^2 \\ &= k_1^2 + k_2^2 + 2k_1 \cdot k_2 \\ &= m^2 + m^2 + 2(E_1E_2 - \vec{p}_1 \cdot \vec{p}_2) \\ &= 2(E_1E_2 + m^2 - \sqrt{E_1^2 - m^2}\sqrt{E_2^2 - m^2}\cos \theta_{12}) \end{aligned}$$

- 2.3. When the decay products are photons, we can set  $m = 0$ . Repeat problem 2.2 in the case where  $m = 0$ .

For  $m = 0$ ,  $|\vec{p}_{1,2}| = E_{1,2}$  and  $m_{12}^2 = 2E_1E_2(1 - \cos \theta_{12})$ .

- 2.4. The Higgs boson was first discovered in 2012 via its decay to photons,  $H \rightarrow \gamma\gamma$ . Suppose the photon energies are 50 GeV and 200 GeV. Given that the mass of the Higgs is  $m_H = 125$  GeV, what is the angle between the two photons in degrees?

For  $H \rightarrow \gamma\gamma$ , the invariant mass is

$$\begin{aligned} m_H^2 &= 2E_{\gamma,1}E_{\gamma,2}(1 - \cos \theta_{12}) \\ \frac{m_H^2}{2E_{\gamma,1}E_{\gamma,2}} &= 1 - \cos \theta_{12} \end{aligned}$$

Plugging in  $E_{\gamma,1} = 50$  GeV,  $E_{\gamma,2} = 200$  GeV, and  $m_H = 125$  GeV, we have  $\theta_{12} \approx 77^\circ$ .

## Part 3: Relativistic kinematics at colliders

A tried-and-true technique for making new particles is by smashing other particles together. For example, positive and negative muons can be created by colliding electrons and positrons,  $e^+ + e^- \rightarrow \mu^+ + \mu^-$ . The electron  $e^-$  and positron  $e^+$  have the same mass,  $m_e$ , and the positive muon  $\mu^+$  and negative muon  $\mu^-$  have the same mass  $m_\mu$ . Let's assign 4-vectors to each particle in this collision and write the energy-momentum conservation equation as

$$p_1 + p_2 = k_1 + k_2 \quad (3)$$

- 3.1. In a typical collider, the electron and positron have equal and opposite momenta, so that the lab frame and the CM frame coincide. Let's write  $p_1 = (E/2, \vec{p})$  and  $p_2 = (E/2, -\vec{p})$ . Find  $|\vec{p}|$  in terms of  $E$  and  $m_e$ .

$$|\vec{p}| = \sqrt{(E/2)^2 - m_e^2}$$

- 3.2. Typically, the colliding particles are highly relativistic,  $E \gg m_e$ . Perform a Taylor expansion of  $|\vec{p}|$  from 3.1 above. If  $m_e = 0.5$  MeV and  $E = 100$  GeV, what is the size of the leading-order term in  $m_e/E$ ? This justifies treating highly relativistic electrons and positrons as if they were massless,  $m_e \approx 0$ .

Factoring out an  $E/2$ , we have

$$|\vec{p}| = \frac{E}{2} \sqrt{1 - \frac{4m_e^2}{E^2}} \approx \frac{E}{2} \left( 1 - \frac{2m_e^2}{E^2} \right)$$

The size of the correction is  $2m_e^2/E^2 = 5 \times 10^{-11}$ . So the electron is approximately massless to an accuracy of better than 1 part in 10 billion!

- 3.3. The 4-vector corresponding to the initial state energy-momentum is  $P = p_1 + p_2 = (E, \vec{0})$ . Energy-momentum conservation thus reads  $P = k_1 + k_2$ , which looks just like the decays you studied in Part 1. Use this fact to find the energies of the outgoing muons. *Hint: you should find that the  $\mu^+$  and  $\mu^-$  have the same energy.*

Using the same dot product tricks as part 1, we write the momentum conservation equation as  $P - k_1 = k_2$ , and square both sides to get  $P^2 + k_1^2 - 2P \cdot k_1 = k_2^2$ . Since  $k_1^2 = k_2^2 = m_\mu^2$ , the  $m_\mu^2$  terms cancel from both sides, and we get  $E^2 - 2EE_1 = 0$  where  $E_1$  is the energy of the muon with 4-momentum  $k_1$ . The solution is just  $E_1 = E/2$  (and hence  $E_2 = E/2$  as well), which makes sense: each muon gets half the energy.